Approximate Domain Unlearning for Vision-Language Models

immediate

Abstract

Pre-trained vision-language models (VLMs) such as CLIP generalize broadly across visual domains, yet downstream deployment often demands that specific domains be selectively forgotten—for example, to satisfy data-privacy regulations or to prevent domain-specific leakage. Existing machine-unlearning methods are designed for *class*-level forgetting and do not transfer naturally to the domain setting, where the same semantic class may appear in multiple domain styles (e.g., real photographs versus sketches). We formulate **Approximate Domain Unlearning (ADU)**: given a set of domains to forget, adapt the VLM so that its predictions on forgotten-domain inputs become uninformative while retaining full accuracy on the remaining domains. Two key challenges arise: (i) domain distributions in VLM feature spaces are highly entangled, making it difficult to isolate and erase a single domain; and (ii) a single learnable prompt cannot capture the instance-level domain cues required for fine-grained control. We address these challenges with two complementary components. First, a **Domain Disentangling Loss (DDL)** that jointly minimizes a domain-classification cross-entropy and maximizes pairwise Maximum Mean Discrepancy (MMD) across domains, explicitly separating domain clusters in the latent space. Second, an **Instance-wise Prompt Generator (InstaPG)** that augments deep vision prompts through cross-attention over image-patch features, producing instance-adaptive prompts that capture fine-grained domain cues. These two components are trained together with memorization-preservation and entropy-maximization objectives on retained and forgotten data, respectively. On three multi-domain benchmarks—Office-Home, Mini DomainNet, and ImageNet (real vs. sketch)—ADU consistently and substantially outperforms strong CLIP fine-tuning and unlearning baselines across all numbers of forgotten domains. Notably, the harmonic mean H of memory accuracy and forgetting error improves from 52.59 to 69.96 on Office-Home and from 62.07 to 75.56 on Mini DomainNet when one domain is forgotten.

1 Introduction

Large vision-language models (VLMs) such as CLIP [Radford et al., 2021], ALIGN [Jia et al., 2021], BLIP [Li et al., 2022], and BLIP-2 [Li et al., 2023] have demonstrated remarkable zero-shot and few-shot capabilities across a wide range of visual recognition tasks. Their success stems from pre-training on billions of image-text pairs that span diverse visual domains: natural photographs, sketches, paintings, product images, cartoons, and more. While this breadth is a strength for generalization, it becomes a liability in privacy-sensitive or domain-restricted deployments.

Consider a medical imaging company that trains a VLM on data from multiple hospital sites (domains), then discovers that one site's data was included without proper consent. Data-protection regulations such as the EU's General Data Protection Regulation (GDPR) require that the influence of this data be removed from the model [Cao and Yang, 2015, Ginart et al., 2019]. In this setting, the unit of forgetting is a *domain*, not a class.

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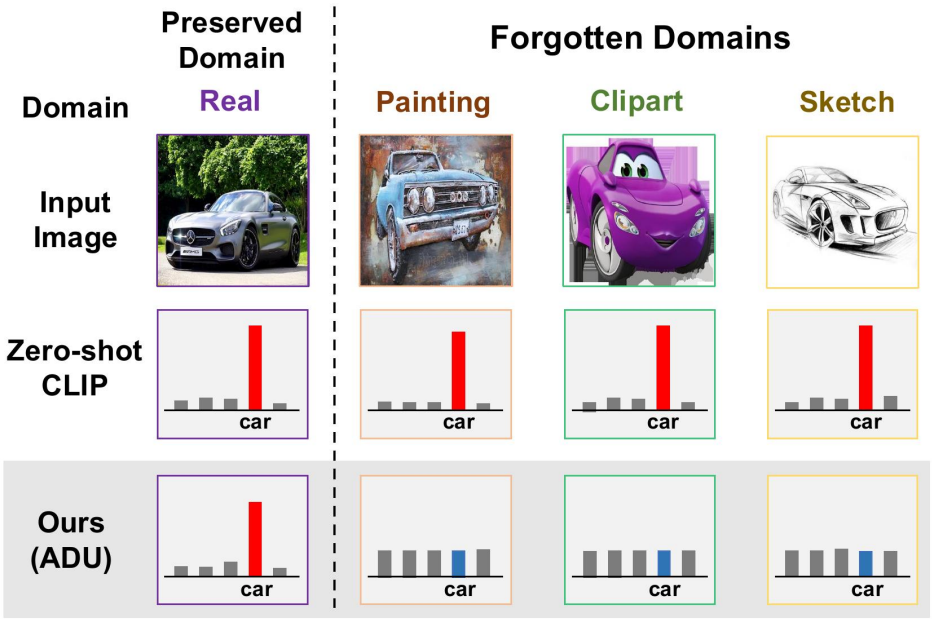


Figure 1: **Illustration of Approximate Domain Unlearning (ADU).** Given a pre-trained VLM and a designated set of domains to forget ($\mathcal{D}_{\text{forget}}$), ADU adapts the model to become uninformative on forgotten-domain inputs while retaining accuracy on the remaining domains ($\mathcal{D}_{\text{retain}}$). Unlike class-level unlearning, domain-level forgetting must operate across all semantic classes simultaneously.

Machine unlearning aims to remove specific training data's influence from a trained model without full retraining [Cao and Yang, 2015, Bourtoule et al., 2021, Xu et al., 2024]. Exact unlearning guarantees are computationally intractable for large neural networks [Sekhari et al., 2021], motivating *approximate unlearning*, which relaxes exact removal guarantees while preserving model utility [Izzo et al., 2021, Golatkar et al., 2020a, Tarun et al., 2023]. Most existing methods focus on forgetting individual data points [Graves et al., 2021, Golatkar et al., 2020b] or entire classes [Chundawat et al., 2023, Fan et al., 2024, Chen et al., 2023], and are designed for discriminative models rather than VLMs.

Domain-level forgetting is fundamentally different from class-level forgetting. A class can often be erased by suppressing the relevant output logit. A domain, by contrast, is an orthogonal dimension of variation: the class "car" appears in real photographs, sketches, paintings, and clipart alike. Forgetting the sketch domain means that the model should become uninformative on *all* sketch-domain inputs, regardless of class—while still correctly recognizing cars in real photographs. Naïve loss reweighting fails here because domain distributions are highly entangled in VLM feature spaces: sketch cars and real cars are mapped nearby by CLIP because they share semantic content.

In this paper, we formalize **Approximate Domain Unlearning (ADU)** for VLMs and propose a method that addresses the entanglement problem through two novel components (Figure 1). The **Domain Disentangling Loss (DDL)** drives the model to explicitly separate domain distributions in feature space by combining a domain-classification cross-entropy (to build domain-discriminative representations) with a term that maximizes pairwise inter-domain MMD [Gretton et al., 2012] (to push domain clusters apart). The **Instance-wise Prompt Generator (InstaPG)** conditions deep vision prompts on the patch-level features of each individual input image via cross-attention, enabling the model to detect and respond to domain cues at the instance level.

We instantiate ADU on CLIP ViT-B/16 with deep vision prompting and evaluate on three benchmarks: ImageNet (real vs. sketch), Office-Home (art / clipart / product / real), and Mini DomainNet (clipart / painting / real / sketch). Across all settings, ADU achieves substantial gains in the harmonic mean H of domain-retention accuracy (*Mem*) and domain-forgetting error (*For*) compared to strong CLIP fine-tuning and machine-unlearning baselines.

Contributions.

- We formulate Approximate Domain Unlearning (ADU), a new problem setting that moves beyond class-level unlearning to the domain dimension in VLMs.
- We propose Domain Disentangling Loss (DDL), which combines CE-based domain classification with MMD maximization to explicitly separate domain distributions for controllable forgetting.
- We introduce Instance-wise Prompt Generator (InstaPG), which leverages cross-attention over image patches to produce instance-adaptive vision prompts that capture fine-grained domain cues.
- We demonstrate consistent and substantial improvements over strong baselines on four multi-domain benchmarks, with robustness to hyperparameters and data-shot settings.

2 Related Work

Vision-Language Models and Prompt Learning. CLIP [Radford et al., 2021] and similar VLMs [Jia et al., 2021, Li et al., 2022, 2023, Yu et al., 2022, Zhai et al., 2023] learn joint image-text representations via contrastive pre-training, enabling zero-shot transfer through natural-language prompts. Context Optimization (CoOp) [Zhou et al., 2022b] introduced learnable text prompt tokens for downstream adaptation, followed by conditional variants (CoCoOp) and multi-modal prompt learning (MaPLe) [Khattak et al., 2023], which jointly learns prompts in both vision and language branches. Test-time prompt tuning [Shu et al., 2022] adapts prompts at inference using entropy minimization. These methods focus on *improving* generalization across domains, whereas ADU seeks to *selectively suppress* specific domains. CLIP-based domain generalization methods [Singha et al., 2023, Huang et al., 2023, Wang et al., 2024, Cho et al., 2023, Yu et al., 2024] adapt VLMs to handle unseen target domains—an orthogonal goal to forgetting seen domains.

Machine Unlearning. Machine unlearning [Cao and Yang, 2015] aims to remove the effect of specific training data from a model, motivated by privacy regulations [Fredrikson et al., 2015, Shokri et al., 2017] and the right to be forgotten. Exact unlearning via retraining is computationally prohibitive [Bourtoule et al., 2021, Sekhari et al., 2021], motivating approximate unlearning [Izzo et al., 2021, Golatkar et al., 2020a,b, 2021, Nguyen et al., 2020, Guo et al., 2020]. Gradient-based methods include weight scrubbing [Graves et al., 2021], boundary shifting [Chen et al., 2023], saliency-guided forgetting [Fan et al., 2024], and teacher-student unlearning [Chundawat et al., 2023]. Forgetting with random forests [Brophy and Lowd, 2021, Sun et al., 2023] and differential privacy-based approaches [Abadi et al., 2016, Neel et al., 2021, Ullah et al., 2021] have also been proposed. Surveys [Shaik et al., 2024, Xu et al., 2024] provide comprehensive taxonomies. Unlearning of features and labels (rather than data points) has been explored [Warnecke et al., 2023], as has recoverable forgetting [Ye et al., 2022] and selective class forgetting [Shibata et al., 2021]. However, none of these works addresses *domain-level* forgetting for VLMs with entangled feature spaces.

Domain Adaptation and Generalization. Domain adaptation transfers models from a source domain to a target domain [Patel et al., 2015, Ganin et al., 2016, Long et al., 2015, 2017, Yan et al., 2017]. Domain generalization trains models that transfer to unseen domains [Zhou et al., 2022a, Li et al., 2018b, D'Innocente and Caputo, 2018, Li et al., 2018a, Mitsuzumi et al., 2021]. Test-time domain adaptation [Wu et al., 2024] adapts at inference. MMD [Gretton et al., 2012] has been widely used to align domain distributions [Long et al., 2015, Yan et al., 2017, Li et al., 2018b]; our DDL instead *maximizes* MMD to disentangle domains for selective forgetting. Concurrent work [Kuwana et al., 2024] considers black-box forgetting in domain-shift settings, and [Huang et al., 2024] proposes a strong linear probe for few-shot CLIP adaptation. CLIPFit [Li et al., 2024] adapts CLIP by fine-tuning specific vision parameters. We compare against all of these as baselines.

3 Method

3.1 Problem Formulation

Let $\mathcal{M}$ be a pre-trained VLM (specifically CLIP ViT-B/16) and let $\mathcal{D} = \{d_1, \ldots, d_K\}$ be the set of visual domains present in the training data. Each data point (x, y, d) consists of an image x, a class label $y \in \mathcal{Y}$, and a domain label $d \in \mathcal{D}$.

Given a designated *forget* set $\mathcal{D}_{\text{forget}} \subset \mathcal{D}$ and a corresponding *retain* set $\mathcal{D}_{\text{retain}} = \mathcal{D} \setminus \mathcal{D}_{\text{forget}}$, the goal of **Approximate Domain Unlearning (ADU)** is to produce an adapted model $\mathcal{M}^*$ satisfying:

1. **Forgetting:** $\mathcal{M}^*$ makes high-entropy (uninformative) predictions on images from $\mathcal{D}_{\text{forget}}$, regardless of class.
2. **Retention:** $\mathcal{M}^*$ maintains high classification accuracy on images from $\mathcal{D}_{\text{retain}}$.

We measure performance via three metrics: *Mem* (accuracy on retained-domain data), *For* (error rate—i.e., 100 minus accuracy—on forgotten-domain data; higher is better), and $H = \frac{2 \cdot Mem \cdot For}{Mem + For}$ (harmonic mean; higher is better).

Following the VLM prompt-learning paradigm, we keep the CLIP backbone frozen and learn only: (i) deep vision prompts (VPT-style tokens inserted at multiple transformer layers), (ii) text prompt tokens (CoOp-style), (iii) a lightweight domain classifier head, and (iv) the cross-attention weights of InstaPG.

3.2 Base Unlearning Objectives

Let $\mathcal{B}_{\text{ret}}$ and $\mathcal{B}_{\text{for}}$ denote mini-batch samples from $\mathcal{D}_{\text{retain}}$ and $\mathcal{D}_{\text{forget}}$, respectively. The base unlearning objectives are:

Memorize loss (retain accuracy on $\mathcal{D}_{\text{retain}}$):

$$\mathcal{L}_{\text{memorize}} = \frac{1}{|\mathcal{B}_{\text{ret}}|} \sum_{(x,y) \in \mathcal{B}_{\text{ret}}} \text{CE}(f_{\mathcal{M}^*}(x),\, y)\,, \tag{1}$$

where CE denotes cross-entropy and $f_{\mathcal{M}^*}(x)$ are the classification logits.

Forget loss (maximize prediction entropy on $\mathcal{D}_{\text{forget}}$):

$$\mathcal{L}_{\text{forget}} = -\frac{1}{|\mathcal{B}_{\text{for}}|} \sum_{x \in \mathcal{B}_{\text{for}}} \text{H}(f_{\mathcal{M}^*}(x))\,, \tag{2}$$

where $\text{H}(p) = -\sum_c p_c \log p_c$ is the Shannon entropy. Minimizing $\mathcal{L}_{\text{forget}}$ maximizes the prediction entropy on forgotten-domain inputs, making the model uninformative for those inputs. The baseline method (referred to as *Baseline* in experiments) trains using only $\mathcal{L}_{\text{memorize}} + \mathcal{L}_{\text{forget}}$ [Shibata et al., 2021].

3.3 Domain Disentangling Loss (DDL)

A core challenge is that domain distributions are heavily entangled in CLIP's feature space (Figure 3). Even though different domains share semantic content, naïve entropy-based forgetting erodes retained-domain performance because the forgotten-domain and retained-domain features are not well-separated.

DDL explicitly *separates* domain clusters in the feature space by training a lightweight domain-classifier head g_ϕ (a single linear layer) and coupling it with an MMD-maximizing objective. Let $z_i = \phi_{\mathcal{M}^*}(x_i)$ denote the image features produced by the adapted model. For a mini-batch with domain labels $\{d_i\}$, DDL is defined as:

$$\mathcal{L}_{\text{DDL}} = \underbrace{\gamma \cdot \text{CE}(g_\phi(z), d)}_{\text{domain CE}} - \underbrace{\lambda \cdot \overline{\text{MMD}}(z, d)}_{\text{MMD maximization}}, \tag{3}$$

where $\gamma, \lambda > 0$ are weighting hyperparameters, and

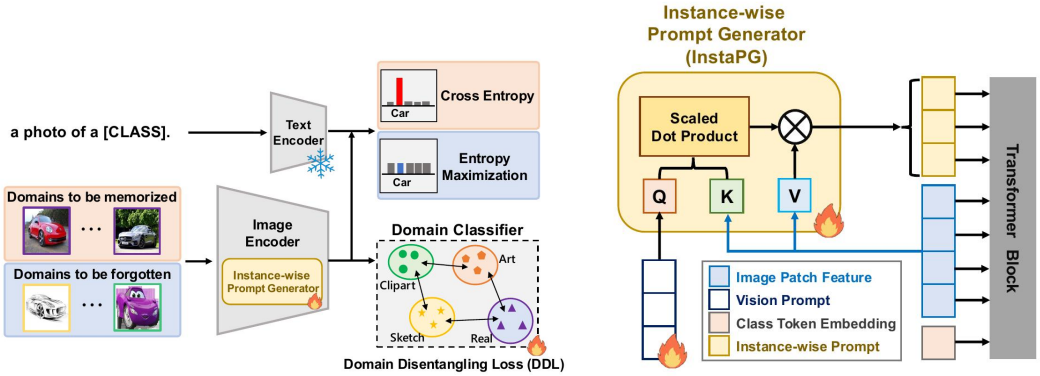
$$\overline{\text{MMD}}(z, d) = \frac{1}{\binom{K}{2}} \sum_{i<j} \text{MMD}^2(\{z : d = d_i\},\, \{z : d = d_j\})\,. \tag{4}$$

We use the Gaussian (RBF) kernel for MMD [Gretton et al., 2012], with bandwidth σ set to the median pairwise distance between domain feature sets [Gretton et al., 2012].

The *domain CE* term trains g_ϕ to correctly classify domains, building discriminative domain representations. The *MMD maximization* term pushes domain distributions apart in the feature space. Together, they create a geometry in which the forgotten domain occupies a well-separated region, making entropy maximization more precise and preventing interference with retained-domain features.

The total training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{memorize}} + \mathcal{L}_{\text{forget}} + \mathcal{L}_{\text{DDL}}. \tag{5}$$



(a) Overview of ADU. (b) Instance-wise Prompt Generator (InstaPG).

Figure 2: **Overview of Proposed Method.** (a) The full ADU framework: frozen CLIP processes each image, and the adapted model generates instance-specific prompts via InstaPG while DDL enforces domain separation. (b) InstaPG uses cross-attention between learnable prompt tokens (queries) and image patch features (keys/values) to produce image-conditioned deep vision prompts.

3.4 Instance-wise Prompt Generator (InstaPG)

Even with well-separated domain distributions, a single shared set of vision prompts may fail to capture fine-grained, instance-level domain cues (e.g., the distinctive texture of a sketch versus the lighting of a real photograph). We propose InstaPG, which generates *instance-adaptive* vision prompts for each image.

As illustrated in Figure 2(b), InstaPG is inserted into the deep vision prompting framework. At each targeted transformer layer ℓ, let $P_\ell \in \mathbb{R}^{M\times d}$ be the M learnable base prompt tokens and let $F_\ell \in \mathbb{R}^{N_p\times d}$ be the N_p patch-token features at that layer. InstaPG augments the prompt tokens via cross-attention:

$$\tilde{P}_\ell = P_\ell + \mathrm{CrossAttn}(Q = P_\ell,\ K = F_\ell,\ V = F_\ell), \tag{6}$$

where $\mathrm{CrossAttn}$ is a lightweight multi-head cross-attention module with learnable projection matrices. The augmented prompt $\tilde{P}_\ell$ is then concatenated with the patch tokens and the class token before being processed by the ℓ-th transformer block. This conditioning on patch-level features allows the prompt to respond to local visual statistics—texture, stroke patterns, rendering artifacts—that are domain-discriminative at the instance level, enabling fine-grained domain control without modifying any CLIP backbone weights.

3.5 Implementation Details

We use CLIP ViT-B/16 [Radford et al., 2021, Dosovitskiy et al., 2021] as the backbone. Deep vision prompts of depth 9 are inserted at layers 1–9 of the vision transformer, with $M = 4$ prompt tokens per layer. Text prompts follow the CoOp/MaPLe design with depth 9 and 4 learnable context tokens [Zhou et al., 2022b, Khattak et al., 2023]. InstaPG cross-attention is applied at all prompt layers. We train for 20 epochs with the SGD optimizer, learning rate 0.002, and cosine annealing. Loss weights are set to $\gamma = 1.0$ and $\lambda = 1.0$ (see sensitivity analysis in Section 4). Training uses 10 shots per domain (class-balanced), consistent with few-shot VLM prompt learning.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on four benchmark settings:

- **ImageNet** [Deng et al., 2009, Wang et al., 2019]: Two domains—real photographs and sketches. Only $|\mathcal{D}_{\text{forget}}| = 1$ is tested (forget one domain, retain the other).

Table 1: **Comparison with Baseline CLIP Fine-tuning Methods.** The average performance of all possible combinations of the domains to be forgotten/retained are reported for various number of domains to be forgotten $|\mathcal{D}_{\text{forget}}| \in \{1, 2, 3\}$. ImageNet has only two domains, so only the case $|\mathcal{D}_{\text{forget}}| = 1$ is tested. Performance is evaluated using the three metrics: (i) *Mem*: the accuracy of the classes for data from memorized domains, (ii) *For*: the error of the classes for data from forgotten domains, and (iii) H: the harmonic mean of *Mem* and *For*. Higher values mean better performance. The *Baseline* in the table refers to the method that trains the vision prompt using $\mathcal{L}_{\text{memorize}}$ and $\mathcal{L}_{\text{forget}}$ in Sec. 3.2, representing a straightforward and conventional approach to unlearning.

$\lvert\mathcal{D}_{\text{forget}}\rvert$	Method	ImageNet			Office-Home			Mini DomainNet		
		$H\uparrow$	*Mem*↑	*For*↑	$H\uparrow$	*Mem*↑	*For*↑	$H\uparrow$	*Mem*↑	*For*↑
$\lvert\mathcal{D}_{\text{forget}}\rvert = 1$	LP++ [Huang et al., 2024]	50.69	62.22	42.76	30.46	**85.06**	18.55	31.73	84.72	19.52
	CLIPFit [Li et al., 2024]	71.31	63.04	82.13	43.44	83.20	29.40	53.56	84.32	39.24
	BBF [Kuwana et al., 2024]	45.56	35.92	65.91	31.25	80.94	19.74	32.12	81.80	20.03
	Baseline	74.66	61.23	96.43	52.59	79.96	39.88	62.07	**85.15**	49.40
	Ours	**77.02**	**64.13**	**96.73**	**69.96**	77.93	**64.34**	**75.56**	78.32	**73.06**
$\lvert\mathcal{D}_{\text{forget}}\rvert = 2$	LP++ [Huang et al., 2024]	×	×	×	31.11	**85.54**	19.01	32.21	**85.29**	19.85
	CLIPFit [Li et al., 2024]	×	×	×	40.53	83.64	26.74	51.35	84.74	36.83
	BBF [Kuwana et al., 2024]	×	×	×	32.54	80.42	20.54	47.95	61.48	40.52
	Baseline	×	×	×	54.19	80.23	41.23	61.97	84.71	48.96
	Ours	×	×	×	**73.58**	75.61	**71.89**	**77.03**	76.51	**77.66**
$\lvert\mathcal{D}_{\text{forget}}\rvert = 3$	LP++ [Huang et al., 2024]	×	×	×	33.57	**86.22**	20.84	34.73	**85.87**	21.77
	CLIPFit [Li et al., 2024]	×	×	×	50.02	85.68	35.32	52.88	85.39	38.30
	BBF [Kuwana et al., 2024]	×	×	×	39.60	78.13	26.80	39.29	80.29	26.05
	Baseline	×	×	×	59.47	80.96	48.79	68.82	84.25	58.27
	Ours	×	×	×	**75.89**	72.15	**80.77**	**79.00**	75.00	**83.94**

- **Office-Home** [Venkateswara et al., 2017]: Four domains—Art, Clipart, Product, and Real World—with 65 object categories. We test $|\mathcal{D}_{\text{forget}}| \in \{1, 2, 3\}$.
- **Mini DomainNet** [Zhou et al., 2021]: A subset of DomainNet [Peng et al., 2019] with four domains—Clipart, Painting, Real, and Sketch—covering 126 classes. We test $|\mathcal{D}_{\text{forget}}| \in \{1, 2, 3\}$.

We report averages over all possible combinations of domains to be forgotten/retained.

Baselines.

- **LP++** [Huang et al., 2024]: Linear probing on CLIP features with strong regularization.
- **CLIPFit** [Li et al., 2024]: Adapts CLIP by fine-tuning selected vision-encoder parameters.
- **BBF** [Kuwana et al., 2024]: Black-box domain forgetting via output-space manipulation.
- **Baseline**: Vision prompt learning using only $\mathcal{L}_{\text{memorize}}$ and $\mathcal{L}_{\text{forget}}$ (Section 3.2).

4.2 Main Results

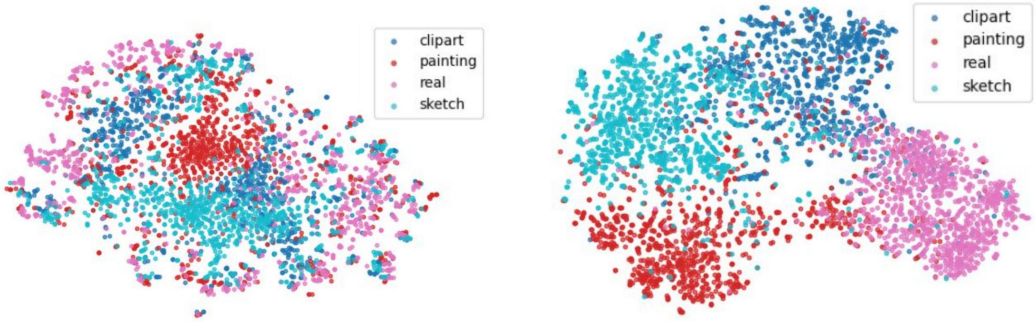
Table 1 reports the main comparison. ADU (Ours) achieves consistent and substantial improvements over all baselines across all datasets and forgetting settings.

On **ImageNet** ($|\mathcal{D}_{\text{forget}}| = 1$), ADU achieves $H = 77.02$ vs. 74.66 for the Baseline, with both improved memorization (*Mem*: 64.13 vs. 61.23) and forgetting (*For*: 96.73 vs. 96.43).

On **Office-Home**, ADU shows dramatic improvements in H: from 52.59 to 69.96 ($|\mathcal{D}_{\text{forget}}| = 1$), from 54.19 to 73.58 ($|\mathcal{D}_{\text{forget}}| = 2$), and from 59.47 to 75.89 ($|\mathcal{D}_{\text{forget}}| = 3$). The *For* metric improves from 39.88 to 64.34, from 41.23 to 71.89, and from 48.79 to 80.77, respectively—demonstrating that ADU is far more effective at erasing forgotten-domain knowledge.

On **Mini DomainNet**, H improves from 62.07 to 75.56 ($|\mathcal{D}_{\text{forget}}| = 1$), from 61.97 to 77.03 ($|\mathcal{D}_{\text{forget}}| = 2$), and from 68.82 to 79.00 ($|\mathcal{D}_{\text{forget}}| = 3$).

While the Baseline and LP++ achieve high *Mem* scores by tending to preserve all domain representations, their *For* scores are much lower, indicating that the forgotten domain is not effectively



(a) Zero-shot CLIP (before unlearning).

(b) ADU (ours), domain to forget: *Real*.

Figure 3: **t-SNE Visualization of Feature Distributions (Office-Home, forget *Real*).** Before unlearning (left), domain distributions are heavily overlapping. After ADU (right), the *Real* domain (to be forgotten) is well-separated from the retained domains, enabling more precise forgetting without interference.

Table 2: **Domain Classification Accuracy when *Art* is Forgotten on Office-Home.**

Method	Domain Classification Accuracy [%]
Before Unlearning	25.80
Ours w/o DDL and InstaPG	31.06
Ours	79.43

erased. ADU achieves a better balance because DDL explicitly separates the domains, enabling precise entropy maximization on the forgotten domain.

4.3 Domain Separability Analysis

To directly assess domain separability, Table 2 reports domain classification accuracy on Office-Home when the Art domain is forgotten—a proxy for how well-separated domains are in the feature space.

Before unlearning, domain classification accuracy is only 25.80%, confirming that CLIP's features are largely domain-agnostic. Without DDL and InstaPG (Baseline), accuracy rises modestly to 31.06%, insufficient for precise domain isolation. With full ADU, domain classification accuracy reaches 79.43%, demonstrating that DDL and InstaPG together create a highly domain-discriminative feature space.

Figure 3 provides a qualitative view: the t-SNE plot for zero-shot CLIP shows heavily overlapping domain clusters, while ADU creates compact, well-separated clusters, with the forgotten *Real* domain clearly isolated.

4.4 Ablation Studies

Effect of DDL and InstaPG. Table 3 ablates the contribution of each component.

On Office-Home ($|\mathcal{D}_{\text{forget}}| = 1$), adding InstaPG alone improves H from 52.59 to 56.41 (with a strong *Mem* boost to 83.55), while adding DDL alone improves H to 60.82 with a substantial *For* gain (51.72 vs. 39.88). The full combination of both components achieves the best H of 69.96, with both *Mem* (77.93) and *For* (64.34) improving over the baseline. This pattern is consistent across all $|\mathcal{D}_{\text{forget}}|$ settings and on Mini DomainNet, confirming that DDL and InstaPG are complementary: DDL improves domain separability (boosting *For*), while InstaPG improves instance-level domain capture (boosting both *Mem* and *For*).

DDL loss component analysis. Table 4 further ablates the two terms within DDL on Office-Home (with InstaPG active).

Table 3: **Ablation Study.** The ablation results of Domain Disentangling Loss (DDL) and Instance-wise Prompt Generator (InstaPG) are reported. While our method achieves improvements with either DDL or InstaPG alone, combining both leads to even better balanced trade-off performance.

Dataset	DDL	InstaPG	$\lvert\mathcal{D}_{\text{forget}}\rvert = 1$			$\lvert\mathcal{D}_{\text{forget}}\rvert = 2$			$\lvert\mathcal{D}_{\text{forget}}\rvert = 3$		
			$H\uparrow$	*Mem*↑	*For*↑	$H\uparrow$	*Mem*↑	*For*↑	$H\uparrow$	*Mem*↑	*For*↑
Office-Home	-	-	52.59	79.96	39.88	54.19	80.23	41.23	59.47	80.96	48.79
	-	✓	56.41	**83.55**	44.05	61.01	**83.33**	48.60	70.60	**81.55**	63.28
	✓	-	60.82	74.51	51.72	60.01	76.22	49.94	63.12	77.74	54.57
	✓	✓	**69.96**	77.93	**64.34**	**73.58**	75.61	**71.89**	**75.89**	72.15	**80.77**
Mini DomainNet	-	-	62.07	85.15	49.40	61.97	84.71	48.96	68.82	**84.25**	58.27
	-	✓	64.06	**85.46**	51.64	65.92	**85.12**	53.96	74.17	83.68	66.68
	✓	-	74.23	78.00	70.87	75.28	77.57	73.20	77.60	77.42	77.90
	✓	✓	**75.56**	78.32	**73.06**	**77.03**	76.51	**77.66**	**81.78**	77.96	**86.12**

Table 4: **Ablation Studies of Loss Functions in Domain Disentangling Loss (DDL).** DDL employs two loss functions: Cross-Entropy (CE) and Maximum Mean Discrepancy (MMD). Using both losses together yields the best-balanced trade-off, as shown by H.

CE	MMD	$\lvert\mathcal{D}_{\text{forget}}\rvert = 1$			$\lvert\mathcal{D}_{\text{forget}}\rvert = 2$			$\lvert\mathcal{D}_{\text{forget}}\rvert = 3$		
		$H\uparrow$	$Mem\uparrow$	$For\uparrow$	$H\uparrow$	$Mem\uparrow$	$For\uparrow$	$H\uparrow$	$Mem\uparrow$	$For\uparrow$
-	-	56.41	**83.55**	44.05	61.01	**83.33**	48.60	70.60	81.55	63.28
-	✓	68.62	82.41	59.47	72.21	81.01	65.66	66.76	**82.38**	57.12
✓	-	64.01	82.97	53.62	69.33	80.73	61.24	71.53	81.43	64.55
✓	✓	**69.96**	77.93	**64.34**	**73.58**	75.61	**71.89**	**75.89**	72.15	**80.77**

Using only MMD maximization (H: 68.62 for $\lvert\mathcal{D}_{\text{forget}}\rvert = 1$) or only domain CE ($H$: 64.01) both outperform the Baseline with InstaPG only (H: 56.41), confirming that each term of DDL independently contributes to domain separation. The full DDL (CE + MMD) achieves the best H of 69.96 by combining the discriminative power of domain classification with the distributional separation afforded by MMD maximization.

4.5 Attention Visualization

Figure 4 visualizes the attention maps of InstaPG on Office-Home images. The cross-attention mechanism focuses on domain-discriminative regions: stroke patterns and artistic textures for the Art domain, clean product surfaces for Product, and photographic details for Real World. This demonstrates that InstaPG captures domain-specific cues at the instance level, rather than relying on class-level or global image statistics.

4.6 Sensitivity Analysis

Loss weight sensitivity. Figures 5a and 5b show the sensitivity of ADU to the DDL weights γ (domain CE weight) and λ (MMD weight). Performance remains stable across a broad range of values for both hyperparameters, indicating that ADU is not highly sensitive to these choices and does not require extensive tuning.

Data efficiency. Figure 6 shows the performance of ADU and the Baseline as a function of the number of training shots per domain (1 to 16 shots). ADU consistently outperforms the Baseline across all shot settings on all three benchmarks, with the advantage maintained even at 1 shot. This demonstrates that ADU's domain disentanglement and instance-adaptive prompting are beneficial even in extremely low-data regimes.

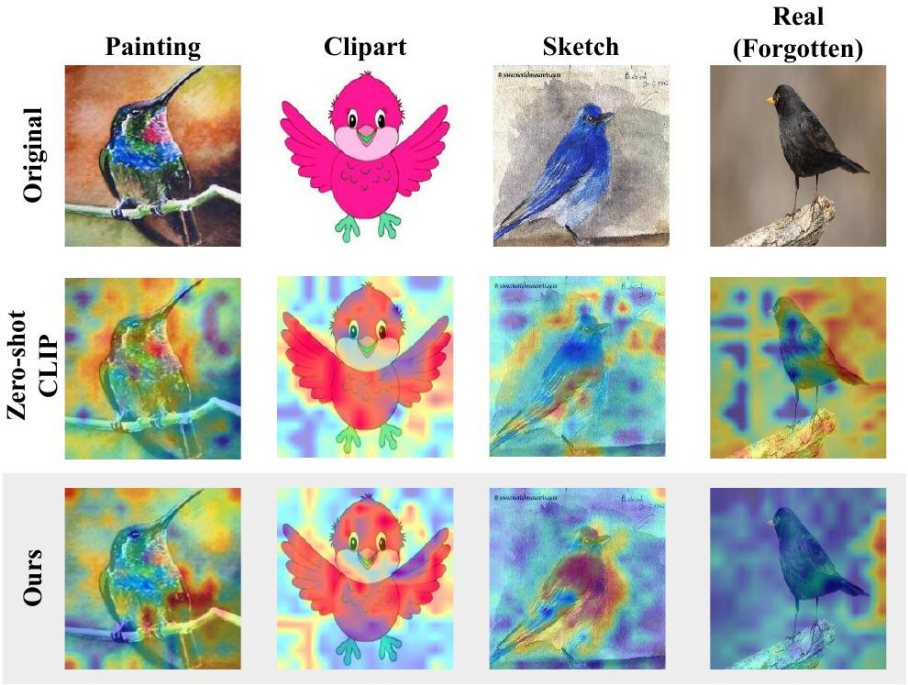
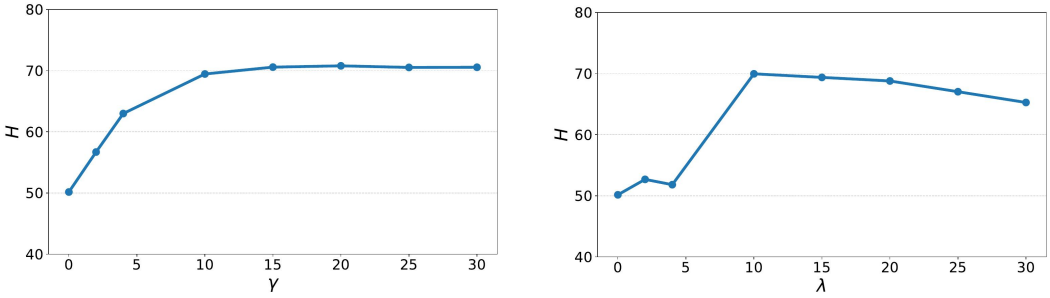


Figure 4: **Visualization of Attention Maps.** InstaPG cross-attention highlights domain-discriminative regions in the input image. The model focuses on texture and style regions that are characteristic of each domain (e.g., stroke patterns for Art, uniform surfaces for Product), enabling fine-grained domain control.



(a) Sensitivity to γ (domain CE weight).

(b) Sensitivity to λ (MMD weight).

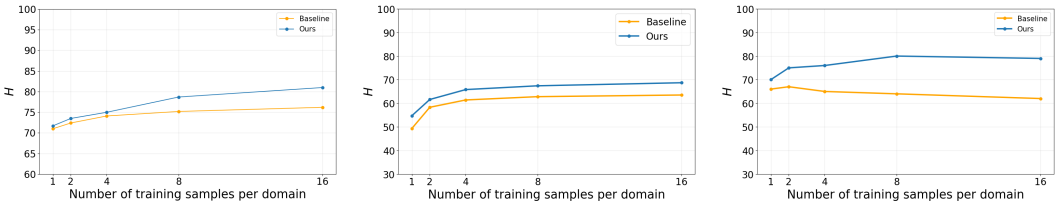
Figure 5: **Impact of Loss Weights γ and λ in DDL.** Performance is stable over a broad range of both hyperparameters, confirming the robustness of the proposed DDL formulation.

5 Limitations

While ADU demonstrates strong performance across multiple benchmarks, several limitations remain.

First, the effectiveness of domain forgetting depends on having a sufficient number of domain-labeled training samples for the forget and retain sets. In truly zero-shot scenarios where no samples from the target forget domain are available, the approach as presented cannot be applied directly.

Second, DDL requires training a domain-classifier head, which in turn requires domain labels for the training data. While domain labels are often available in structured multi-domain datasets, they may not be readily available in general deployment scenarios.



(a) ImageNet. (b) Office-Home. (c) Mini DomainNet.

Figure 6: **Sensitivity to the Number of Training Samples per Domain.** ADU consistently outperforms the Baseline across all shot settings, and advantages are maintained even with very few training samples.

Third, while ADU demonstrates robustness across the tested benchmarks (ImageNet, Office-Home, Mini DomainNet), its effectiveness on VLMs with significantly different architectures or pre-training objectives (e.g., decoder-based generative VLMs) has not been validated. Additionally, the formal privacy guarantee of the unlearning process—how well the adapted model aligns with an exactly retrained model—has not been quantitatively characterized, which is important for regulatory compliance.

6 Conclusions

We have presented Approximate Domain Unlearning (ADU), a new framework for selectively forgetting visual domains in pre-trained vision-language models. ADU addresses the core challenge of entangled domain distributions through two complementary components: Domain Disentangling Loss (DDL), which combines domain-classification cross-entropy with inter-domain MMD maximization to explicitly separate domain clusters in feature space, and Instance-wise Prompt Generator (InstaPG), which conditions deep vision prompts on image-patch features via cross-attention to capture instance-level domain cues.

Extensive experiments on Office-Home, Mini DomainNet, and ImageNet demonstrate that ADU consistently and substantially outperforms strong CLIP fine-tuning and machine-unlearning baselines across all forgetting settings. Ablation studies confirm that both DDL and InstaPG contribute independently, and that their combination is essential for optimal performance. ADU achieves high domain separability (79.43% domain classification accuracy vs. 25.80% before unlearning) and robust performance across different numbers of forgotten domains, data-shot settings, and hyperparameter configurations.

This work opens a new direction at the intersection of machine unlearning and vision-language model adaptation, highlighting the distinct challenges that arise when the unit of forgetting is a visual domain rather than a class. Future work could explore extending ADU to generative VLMs, developing formal privacy guarantees, and enabling forgetting without domain-labeled data.

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